

Generation Interconnection System Impact Study Report

For Queue Project Q15-0432

Shady Oaks Wind Farm 500 kV

August 25, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0432. The Interconnection Customer proposes a 1083.8 MW natural gas combined cycle facility interconnecting at the Shady Oaks Wind Farm 500 kV substation in the Illinois area. The request was received on February 07, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 216 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$80,883,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Shady Oaks Wind Farm Substation 500/230kV Auto	Shady Oaks Wind Farm to Alder 500.0 kV line	173.8	1,183	100.0
Shady Oaks Wind Farm to Alder Substation 230.0 kV	Shady Oaks Wind Farm to Alder 500.0 kV	171.5	856	67.5
Ivyglen Meter to Royalton 230.0 kV	Ft St Vrain to Alder 500.0 kV	110.0	525	15.6
Ivyglen Meter to Alder 230.0 kV Circuit 1	Ft St Vrain to Alder 500.0 kV	102.8	718	16.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Shady Oaks Wind Farm 500-230 kV XFMR	Replace transformer with higher-rated unit	\$35,008,000	100% (\$35,008,000)
Shady Oaks Wind Farm to Alder 230.0 kV	Rebuild line with higher-capacity conductor	\$31,871,000	99% (\$31,500,000)
Ivyglen Meter Substation to Royalton 230.0 kV Line	Reconductor line	\$5,254,000	52% (\$2,752,000)
Ivyglen Meter to Alder 230.0 kV Ckt 1	Reconductor line	\$19,959,000	58% (\$11,623,000)
	Total allocated to project		\$80,883,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.